Enterprise Lead Management Master Blueprint Appendix – Reference Designs & Implementation Assets

A. End-to-End Reference Flow

Sales Representative → WhatsApp Business → Meta WhatsApp API → Automation Trigger (n8n/Composio) → Pipeline Orchestrator → Claude AI → Validation → Duplicate Detection → Databricks (Bronze → Silver → Gold) → Lead Enrichment → Representative Matching → Notification Engine → Quip Sync → React Dashboard.

B. Sample Representative CSV

RepresentativeID, RepresentativeName, Territory, Region, ProductCategory, Language, Manager, ManagerPhone, Email, Workload, ActiveFlag

C. Sample WhatsApp Messages

New Lead: Company: ABC Trading Contact: Ahmed Ali Phone: +974... Product: Tissue Paper Quantity: 300 Status Update: LeadID: L-10025 Status: Quote Sent Remarks: Follow-up Friday.

D. Suggested Delta Tables

leads, customers, representatives, managers, products, lead_activity, lead_status_history, notifications, reminder_queue, audit_log, dashboard_metrics, configuration.

E. API Examples

POST /api/leads PUT /api/leads/{id}/status POST /api/representatives/upload-csv GET /api/dashboard GET /api/notifications

F. Frontend Folder Structure

src/ components/ pages/ layouts/ services/ api/ hooks/ store/ types/ utils/ assets/ styles/

G. Backend Folder Structure

app/ api/ services/ orchestrator/ integrations/ repositories/ models/ schemas/ auth/ workers/ tests/

H. Environment Variables

WHATSAPP_API_TOKEN WHATSAPP_WEBHOOK_SECRET DATABRICKS_HOST DATABRICKS_TOKEN DATABRICKS_SQL_WAREHOUSE QUIP_API_TOKEN JWT_SECRET REDIS_URL

I. Dashboard Pages

Executive Landing Lead Management Sales Representative Sales Manager Marketing Analytics Notifications Administration Operations Dashboard

J. Operational KPIs

Lead ingestion latency, AI extraction accuracy, duplicate rate, SLA compliance, reminder success rate, pipeline execution time, dashboard freshness, API latency, notification delivery rate.

K. Future Enhancements

ERP integration, CRM connectors, OCR ingestion, voice transcription, multilingual AI, predictive forecasting, AI Copilot, mobile app, customer portal.

L. Recommended Build Order

1. Databricks foundation 2. WhatsApp integration 3. Pipeline Orchestrator 4. AI extraction 5. Dashboard 6. Notifications 7. Security 8. UAT 9. Production rollout